TO: Head of Fraud Risk, Deacon Financial Services

FROM: Team 1 (Slow Burn): Adan Urizar, Ben Xu, Catherine Burke, Jordan Riggs, Kennedy Kirk

DATE: November 13, 2026

SUBJECT: XGBoost catches 19% more fraud than our HW4 model and lowers fraud-related costs by $103,000 on the test set

1. Executive Summary

We tested a random forest and an XGBoost model against our HW4 logistic regression on the same applications, and we recommend XGBoost, with applications scoring above 0.16 sent to manual review. On the 52,998 test applications it catches 54% of confirmed fraud while reviewing 6.8% of applications, compared with 46% for the logistic model, and lowers total fraud and review cost from $728,093 to $624,928 ($983,963 with no model). The extra review volume fits within the fraud-review team's current capacity, so the change requires no new hires.

2. Objective

DFS decides at submission whether to approve an application or send it to manual review. Our task was to test whether models beyond logistic regression predict confirmed fraud better, compare them with our HW4 model, and recommend one final model and review cut-off.

Each missed fraud costs DFS $305 to $505, depending on whether the $200 New Money Bonus and a referral reward applied. Each legitimate customer sent to review costs about $80 in review time and lost customer value, since about 16% of reviewed applicants abandon. The goal is the lowest combined cost within the volume the review team can handle.

3. Key Findings

- Both tree models outperform the HW4 logistic regression. On the same random 25% test set, test AUC is 0.870 for XGBoost and 0.868 for the random forest, against 0.838 for the logistic model refit on the updated data (0.844 in HW4) and 0.645 for the bank's current score. Reviewing the riskiest 5% of applications finds 46% of all fraud with XGBoost, 40% with the logistic model and 6% with the bank's current score.
- Why we chose XGBoost over the HW4 logistic model: it is better on every business measure at once. On the test set it saves $103,165 (14% lower total cost), catches 238 more fraud cases (54% of fraud against 46%), sends 165 fewer legitimate customers to review, and raises the share of flags that are real fraud from 37% to 43%. We accept that it is harder to explain than the logistic model because every flag still goes to a human reviewer.
- We recommend XGBoost over the random forest. The two are effectively tied on AUC and cost ($624,928 vs. $626,269), but XGBoost flags 203 fewer legitimate customers, shows less overfitting (training AUC 0.879 vs. test 0.870, against 0.922 vs. 0.868 for the forest) and is much faster to train.
- The team's feature engineering drives the result. Features we built from our HW3 ideas carry 92% of XGBoost's importance and nine of its ten strongest inputs; the 28 raw

application fields carry 5%. The strongest group is links to other applications through a shared phone, address, payout account or device (46%), which were unavailable in HW4.

4. Implications & Recommendations

1. Deploy XGBoost and review every application scoring above 0.16. Applied to all August 2026 applications, the cut-off flags about 4,400 per month. The fraud-review teams had about 21,500 cases of scheduled capacity in August and received 3,300, leaving room for roughly 18,000.
2. Keep the cut-off under review. With conservative costs (bonus and onboarding only for a missed fraud, $48 per review), the lowest-cost cut-off rises to 0.25 and cost still falls from $266,014 with no model to $216,576.
3. Accept the loss of interpretability, with safeguards. The logistic model explains each factor with one odds ratio; XGBoost does not. We report which factors drive it and how risk moves with each (feature importance and partial dependence in the notebook), and every flag goes to a human reviewer, so the model never declines a customer on its own.
4. Make the shared-entity links available at scoring time. They are the model's strongest input and are loaded in batch today.
5. Handle stolen-identity and first-party fraud separately. XGBoost catches 74% of bonus abuse and 71% of synthetic identities but only 37% of stolen identities and 18% of first-party fraud, which look normal at application and need identity checks or post-opening monitoring.
6. Retrain monthly while the bonus is live. Fraud rose from 4.8% of applications (January 2025 to July 2026) to 7.9% in August. Track AUC, recall at the cut-off and the share of flags confirmed as fraud.

Risks. About 57% of flagged applicants are legitimate, so review wait times and abandonment should be monitored. Fraud that was never investigated is labeled as legitimate, so true recall is somewhat lower than reported. The 4 hours of investigation per case in the cost estimate is an assumption, since case hours are not recorded.

5. Supporting Analysis

Data and split. We used the November 2026 DFS export (285,046 applications). An application is fraud if an investigation confirmed it. We modeled January 2025 through August 2026 and split those 211,992 applications at random, 75% training and 25% testing, as in HW4 and the course code. September 2026, whose labels are less mature, served as an extra check on the newest month (XGBoost AUC 0.890). Every feature uses only information available before the application was submitted.

Models. The HW4 logistic regression was refit with its same 20 variables and preparation. Both tree models used all 110 features, since trees handle missing values and select variables through their splits. Both were tuned with Bayesian optimization (Optuna, Deck 9) rather than grid search, on a validation slice taken from the training set: 40 trials for XGBoost (final model: 328 trees of depth 2, learning rate 0.12, with L1/L2 penalties) and 20 for the random forest (final model: 600 trees, depth 16, 16 features per split, at least 16 applications per leaf).

Overfitting checks. Five-fold cross-validated AUC on the training set was 0.870 for XGBoost and 0.869 for the random forest, matching their test AUC. In the random-noise test (Deck 9), 64 of the 110 features beat a column of random numbers.

Cut-off. For each model the cut-off was chosen on the training set by minimizing total cost, then applied to the test set. Missed-fraud cost includes the bonus only if the application's offer paid one and the referral reward only if a code was used.

6. Appendix

Table A1. Model comparison, test set (52,998 applications, 5.3% fraud), lowest-cost cut-off

Model	Train AUC	Test AUC	Cut-off	Reviewed	Recall	Precision	F1	Good reviewed	Cost
Logistic (HW4)	0.843	0.838	0.14	6.6%	45.6%	36.6%	0.406	2,224	$728,093
Random forest	0.922	0.868	0.16	7.2%	55.7%	41.0%	0.472	2,262	$626,269
XGBoost*	0.879	0.870	0.16	6.8%	54.0%	42.5%	0.476	2,059	$624,928
No model				0%	0%			0	$983,963

*Recommended. Recall = share of fraud caught; Good reviewed = legitimate customers sent to review.

Table A2. XGBoost cut-off options, test set

Rule	Cut-off	Reviewed	Recall	Precision	F1	Good reviewed	Cost
Youden	0.05	19.8%	75.9%	20.4%	0.321	8,359	$911,602
Lowest cost*	0.16	6.8%	54.0%	42.5%	0.476	2,059	$624,928
Max F1	0.24	5.2%	47.6%	48.3%	0.479	1,435	$638,552

Table A3. Share of each fraud type caught, test set

Fraud type	Cases	HW4 logistic	Random forest	XGBoost
Bonus abuse	833	64%	76%	74%
Synthetic identity	727	47%	72%	71%
Stolen identity	856	39%	40%	37%
First-party	402	20%	18%	18%

Table A4. What drives XGBoost

Feature source	Features	Share of importance	Among ten strongest
Built by the team from HW3 ideas	75	92%	9
Raw application fields, used as given	28	5%	0
Bank's own pre-computed counts	7	3%	1

Strongest groups: shared phone/address/payout/device links (46%), identity and email (20%), application behavior (13%), device (7%). Full output is in the Jupyter notebook.